CÁLCULO DE LÍMITES

LÍMITES INDETERMINADOS

$\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 0^0, ∞^0, 1^∞

Para cada, necesitamos una de las técnicas siguientes:

CÁLCULO DE DESIGUALDADES (TRIÁNGULO)

CASO 1: $\lim_{x \to 0} \frac{\sin(x)}{x} = \lim_{x \to 0} \frac{\sin(x)}{x} \cdot \frac{\cos(x)}{\cos(x)} = \lim_{x \to 0} \frac{\sin(x)\cos(x)}{x\cos(x)} = \lim_{x \to 0} \frac{\sin(2x)}{2x} \cdot \frac{1}{\cos(x)} = 1 \cdot \frac{1}{\cos(0)} = 1 \cdot \frac{1}{1} = 1$

CASO 2: $\lim_{x \to 0} \frac{1-\cos(x)}{x} = \lim_{x \to 0} \frac{1-\cos(x)}{x} \cdot \frac{1+\cos(x)}{1+\cos(x)} = \lim_{x \to 0} \frac{1-\cos^2(x)}{x(1+\cos(x))} = \lim_{x \to 0} \frac{\sin^2(x)}{x(1+\cos(x))} = \lim_{x \to 0} \frac{\sin(x)}{x} \cdot \frac{\sin(x)}{1+\cos(x)} = \lim_{x \to 0} \frac{\sin(x)}{x} \cdot \lim_{x \to 0} \frac{\sin(x)}{1+\cos(x)} = 1 \cdot \frac{\sin(0)}{1+\cos(0)} = 1 \cdot \frac{0}{2} = 0$

$\lim_{x \to 0} \frac{\sin(x)}{x} = 1$

$\lim_{x \to 0} \frac{1-\cos(x)}{x} = 0$

CASO 3: $\lim_{x \to 0} \frac{\sin(x)}{\tan(x)} = \lim_{x \to 0} \frac{\sin(x)}{\frac{\sin(x)}{\cos(x)}} = \lim_{x \to 0} \frac{\sin(x)\cos(x)}{\sin(x)} = \lim_{x \to 0} \cos(x) = \cos(0) = 1$

CASO 4: $\lim_{x \to 0} \frac{\tan(x)}{x} = \lim_{x \to 0} \frac{\frac{\sin(x)}{\cos(x)}}{x} = \lim_{x \to 0} \frac{\sin(x)}{x \cdot \cos(x)} = \lim_{x \to 0} \frac{\sin(x)}{x} \cdot \frac{1}{\cos(x)} = 1 \cdot \frac{1}{\cos(0)} = 1 \cdot \frac{1}{1} = 1$

CASO 5: $\lim_{x \to 0} \frac{\sin(x)}{\sin(x)} = \lim_{x \to 0} 1 = 1$

CASO 6: $\lim_{x \to 0} \frac{\tan(x)}{\tan(x)} = \lim_{x \to 0} 1 = 1$

CASO 7: $\lim_{x \to 0} \frac{\sin(x)}{\tan(x)} = \lim_{x \to 0} \frac{\sin(x)}{\frac{\sin(x)}{\cos(x)}} = \lim_{x \to 0} \frac{\sin(x)\cos(x)}{\sin(x)} = \lim_{x \to 0} \cos(x) = \cos(0) = 1$

CASO 8: $\lim_{x \to 0} \frac{\tan(x)}{\sin(x)} = \lim_{x \to 0} \frac{\frac{\sin(x)}{\cos(x)}}{\sin(x)} = \lim_{x \to 0} \frac{\sin(x)}{\sin(x) \cdot \cos(x)} = \lim_{x \to 0} \frac{1}{\cos(x)} = \frac{1}{\cos(0)} = \frac{1}{1} = 1$

CASO 9: $\lim_{x \to 0} \frac{\sin(x)}{\sin(2x)} = \lim_{x \to 0} \frac{\sin(x)}{2\sin(x)\cos(x)} = \lim_{x \to 0} \frac{1}{2\cos(x)} = \frac{1}{2\cos(0)} = \frac{1}{2 \cdot 1} = \frac{1}{2}$

CASO 10: $\lim_{x \to 0} \frac{\tan(x)}{\tan(2x)} = \lim_{x \to 0} \frac{\frac{\sin(x)}{\cos(x)}}{\frac{\sin(2x)}{\cos(2x)}} = \lim_{x \to 0} \frac{\sin(x)\cos(2x)}{\sin(2x)\cos(x)} = \lim_{x \to 0} \frac{\sin(x)\cos(2x)}{2\sin(x)\cos(x)\cos(x)} = \lim_{x \to 0} \frac{\cos(2x)}{2\cos^2(x)} = \frac{\cos(0)}{2\cos^2(0)} = \frac{1}{2 \cdot 1^2} = \frac{1}{2}$

LÍMITES DE CUOCIENTE

Con ayuda de la gráfica de $\sin(x)$ y $\cos(x)$ se puede ver que $\sin(x) \leq x$ para $x \geq 0$ y $\cos(x) \geq 1$ para $x \geq 0$.

$\lim_{x \to 0} f(x) = \lim_{x \to 0} \frac{\sin(x)}{x} = 1$

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